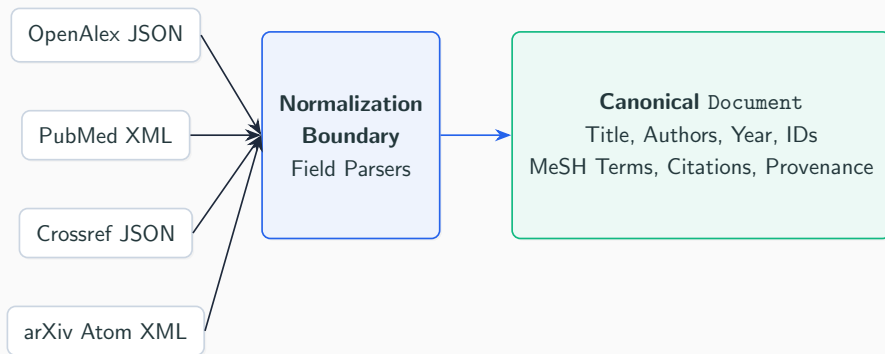


Building a Reproducible Scholarly Search Toolkit

Episode 4: The Canonical Document Model

Nexus Scholar Suite

The Normalization Boundary



The Document Dataclass

```
@dataclass
class Document:
    title: str
    year: int | None = None
    provider: str = "unknown"
    provider_id: str = ""
    external_ids: ExternalIds = field(default_factory=ExternalIds)
    abstract: str | None = None
    authors: list[Author] = field(default_factory=list)
    venue: str | None = None
    citations_count: int | None = None
    mesh_terms: list[str] = field(default_factory=list)
    tldr: str | None = None
    query_id: str | None = None
    retrieved_at: datetime | None = None
```

Scientific Traceability

- `provider_id`: Primary key in remote repository.
- `query_id`: Links to exact search instrument.
- `retrieved_at`: ISO UTC timestamp.

Enriched Metadata

- Medical Subject Headings (`mesh_terms`).
- AI Abstract Summaries (`tldr`).
- Citation Graph Counts (`citations_count`).